Smart AgriTech EMS Documentation

CF Features — End-to-End Test Guide

Purpose: Step-by-step manual QA for all CF features added to Smart AgriTech EMS (web + backend).
Related docs: 09 — CF features implementation
Audience: QA, developers, demo reviewers

Use this guide as a checklist. Mark each case **Pass / Fail / Blocked** and note environment details at the top of your run.

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1. Test run sheet

Field	Value
Tester	
Date	

Field	Value
Build / commit	
Backend URL	e.g. `http://localhost:5000`
Frontend URL	e.g. `http://localhost:5173`
Org under test	
Devices with live telemetry?	Yes / No
Overall result	Pass / Fail

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2. Prerequisites

2.1 Environment

1. PostgreSQL reachable via `DATABASE_URL`
2. Redis optional but recommended (latest metrics / power-flow loads)
3. CF tables applied:

```
cd ems/ems-backend
npx prisma generate
npx prisma db execute --file prisma/add_cf_features.sql
npm run dev
```

4. Frontend:

```
cd web_frontend
npm install
npm run dev
```

5. Ideally ≥ **2 devices** in one org, with recent readings (`ActivePower` / `PowerConsumption`). Optional: `npm run simulate:production` (or your ingest/simulator) so Power Flow and widgets show non-zero values.

2.2 Features covered by this guide

#	Feature	Primary role
A	Access Groups	Org Admin (+ verify as User)
B	Device Groups	Org Admin
C	Facilities + device link	Org Admin

#	Feature	Primary role
D	Power Flow mind map	Org Admin
E	Custom dashboards	Org Admin, User, (Admin)
F	Live widgets / slab cost / refresh	All dashboard roles
G	Device ACL (Access Group ∪ DeviceUser)	User

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3. Accounts & URLs

Default seeded users (confirm against your DB/seed):

Role	Email	Password (typical seed)	Portal
Super Admin	`superadmin@ems.com`	`Admin@123456`	`/admin/...`
Org Admin	`orgadmin@ems.com`	`Admin@123456`	`/org/...`
User	`user@ems.com`	`Admin@123456`	`/user/...`

> If passwords differ in your environment, record them in the test run sheet.

Key UI routes

Feature	Org Admin	Super Admin	User
Access Groups	`/org/access-groups`	`/admin/access-groups`	—
Device Groups	`/org/device-groups`	`/admin/device-groups`	—
Facilities	`/org/settings` (Facility section)	—	—
Power Flow	`/org` (Org Dashboard)	—	—
Custom Dashboards	`/org/custom-dashboard`	`/admin/custom-dashboard`	`/user/custom-dashboard`
Devices	`/org/devices`	`/admin/devices`	Device selectors / detail

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4. Pre-flight checks

ID	Step	Expected	Result
PF-01	Open frontend login	Login page loads	
PF-02	Login as Org Admin	Redirect to `/org` dashboard	
PF-03	Sidebar shows **Access Groups**, **Device Groups**, **Custom Dashboards**	Links present	
PF-04	Backend health / login API works	No 5xx on login	
PF-05	Org has ≥ 1 device	Device list not empty	
PF-06	Open browser Network tab	API base points to correct backend	

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5. Test data setup

Do this once before feature tests. Record names used.

5.1 Identify devices

Alias	Device name	Device ID	Notes
D1			Prefer ONLINE + telemetry
D2			Prefer different from D1
D3 (optional)			For negative ACL tests

5.2 Identify users

Alias	Email	Role
OA	orgadmin@...	ORG_ADMIN
U1	user@...	USER (will get access group)
U2 (optional)	another user	USER **not** in access group

5.3 Suggested naming for this run

Entity	Suggested name
Access group	`E2E Access Lab`
Device group A	`E2E Kitchen Load`
Device group B	`E2E HVAC Load`
Facility building	`E2E Building A`
Facility floor	`E2E Floor 1`
Custom dashboard	`E2E Live Board`

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6. Access Groups E2E

Login as: Org Admin → **Access Groups** (`/org/access-groups`)

6.1 Create

ID	Steps	Expected	Result
AG-01	Click create / add group	Form opens	
AG-02	Enter name `E2E Access Lab`	Name accepted	
AG-03	Select devices **D1** (and optionally D2)	Devices listed as selected	
AG-04	Select user **U1**	User listed as selected	
AG-05	Save	Group appears in list with device/user counts	

6.2 Update

ID	Steps	Expected	Result
AG-06	Edit group; remove D2 if present; keep D1	Save succeeds	
AG-07	Add/remove a user and save	Membership updates on reload	

6.3 Verify as User (ACL — critical)

ID	Steps	Expected	Result
AG-08	Logout; login as **U1**	User portal loads	

ID	Steps	Expected	Result
AG-09	Open Devices / device selectors available to U1	**Only** devices in Access Group (and any DeviceUser assignments) appear	
AG-10	Open a permitted device detail / dashboard charts	Telemetry loads (200)	
AG-11	(If possible) Call or open a device **not** in group	Device missing from list **or** API `403` / `404`	

6.4 Delete

ID	Steps	Expected	Result
AG-12	Login as Org Admin; delete `E2E Access Lab`	Group removed	
AG-13	Login as U1 again (if U1 had no DeviceUser)	Previously group-only devices no longer visible	

> **Tip:** For AG-13, either re-create the group after delete tests, or use a dedicated throwaway group so later tests still have ACL data. Prefer: finish ACL tests, then recreate `E2E Access Lab` for the rest of the day.

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7. Device Groups E2E

Login as: Org Admin → **Device Groups** (`/org/device-groups`)

ID	Steps	Expected	Result
DG-01	Create `E2E Kitchen Load` ; assign **D1**	Group saved	
DG-02	Create `E2E HVAC Load` ; assign **D2** (or D1 if only one device)	Group saved	
DG-03	Edit description / devices; save	Updates persist after refresh	
DG-04	List shows device counts	Counts match selections	
DG-05	Deactivate or delete a throwaway group (optional)	Soft/hard delete behaves as UI states	

Pass criteria: Groups exist for Power Flow tests in §9.

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8. Facilities hierarchy & device linking E2E

Login as: Org Admin → **Settings** (`/org/settings`) → Facility Hierarchy

8.1 Build hierarchy

ID	Steps	Expected	Result
FH-01	Enter Building: `E2E Building A`	Field accepts value	
FH-02	Enter Floor: `E2E Floor 1`	Field accepts value	
FH-03	Optionally Department: `E2E Dept Ops`	Field accepts value	
FH-04	Click **Save Changes** on Facility card	Toast success; tree reloads from API	

8.2 Link devices

ID	Steps	Expected	Result
FH-05	In **Link devices to facilities**, select `E2E Building A` (or Floor)	Node selectable	
FH-06	Check **D1**	Checkbox stays checked	
FH-07	Select Floor node; check **D2** (if available)	Link stored on that node	
FH-08	Save facility section again	After reload, checkboxes still match	

8.3 Consume links in dashboards

ID	Steps	Expected	Result
FH-09	Open Custom Dashboard editor; use facility context filters	Buildings/floors from hierarchy appear	
FH-10	Widget with **group by** facility children	Bars/pie show child names; values reflect linked devices when telemetry exists	

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9. Power Flow mind map E2E

Login as: Org Admin → **Dashboard** (`/org`)

ID	Steps	Expected	Result
PF-01	Scroll to **Power Flow** card	Mind map renders (sources → load → groups)	
PF-02	Confirm device groups appear as branches	`E2E Kitchen Load` / `E2E HVAC Load` visible	
PF-03	With live telemetry + Redis/DB values	Group **kW** not stuck at `0.00` for devices that have `ActivePower`	
PF-04	Source cards show Grid / Solar / Generator	Values present; solar may be 0 if no `ExportPower`	
PF-05	Edit a source value (if editable) and save	Toast/success; reload keeps or re-derives per rules	
PF-06	Click through to device groups path (if link shown)	Navigates to `/org/device-groups`	
PF-07	Refresh page	Power Flow loads again without error	

Blocked if: No telemetry — document as Blocked, still verify UI structure and groups list.

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10. Custom Dashboard Builder E2E

10.1 Org Admin — create & edit

Login as: Org Admin → **Custom Dashboards** (`/org/custom-dashboard`)

ID	Steps	Expected	Result
CD-01	Open list page	Empty state or existing list	
CD-02	Create dashboard `E2E Live Board` ; optionally set default device D1	Dashboard created; editor opens	
CD-03	Add widget: **Stat** / Active Power	Tile on grid	
CD-04	Add widget: **Area** / Energy Consumption	Chart tile	

ID	Steps	Expected	Result
CD-05	Add widget: **Table**	Device rows	
CD-06	Add widget: **Alarms**	Alarm list or empty state	
CD-07	Drag/resize tiles	Layout persists after **Save**	
CD-08	Open widget settings; set target device D1; time range Today	Settings save	
CD-09	Set visibility **Shared** (if available); save	Listed for User later	
CD-10	Reload editor by URL	Layout + widgets restored from API	

10.2 Widget catalog coverage (spot-check)

Add at least one of each major type on a scratch dashboard or the same board:

ID	Widget type	Metric suggestion	Result
CD-11	Line	Voltage	
CD-12	Bar	Current	
CD-13	Pie	Energy or comparison	
CD-14	Gauge	Power Factor	
CD-15	Heatmap	Active Power (needs device)	
CD-16	Text / Markdown	Notes	
CD-17	Multiseries	Power + PF	

10.3 User portal

ID	Steps	Expected	Result
CD-18	Login as User → `/user/custom-dashboard`	Sees own + **Shared** dashboards	
CD-19	Open shared `E2E Live Board`	Widgets render (ACL may limit devices)	
CD-20	Create a **private** user dashboard	Only that user sees it	

ID	Steps	Expected	Result
CD-21	Confirm User **cannot** open Access Groups / Device Groups in nav	Links absent	

10.4 Super Admin spot-check

ID	Steps	Expected	Result
CD-22	Login Super Admin → `/admin/custom-dashboard`	List/create works	
CD-23	Access Groups / Device Groups under `/admin/...`	CRUD works for chosen org (if org scoped)	

10.5 Delete

ID	Steps	Expected	Result
CD-24	Delete a throwaway dashboard	Removed from list	

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11. Live widgets, cost, refresh E2E

Use `E2E Live Board` with target device **D1** that has recent data.

ID	Steps	Expected	Result
LV-01	Stat/Gauge shows numeric value (not forever spinner)	Value or clear empty state	
LV-02	Time series widgets show points for Today / Week / Month	Chart not empty when aggregate exists	
LV-03	Switch widget/dashboard time range to **Year**	Request uses long window (`365d`); chart renders or empty gracefully	
LV-04	Metric **Energy Cost**	Value = energy × slab average (or fallback rate)	
LV-05	Metric **Carbon Emissions**	Scales with energy	
LV-06	Wait ~30 seconds without reload	Widget data refreshes (polling)	
LV-07	With sockets enabled, push new reading	Chart/stat updates without full page reload	

ID	Steps	Expected	Result
LV-08	Clear target device; org-level table/online metrics	Still shows device list based metrics	
LV-09	Group-by facility with linked devices	Comparison values track linked device loads	

How to confirm cost uses slabs: Org/User → Slab Rates; note average rate; compare cost widget ≈ kWh × rate.

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12. ACL & security E2E

ID	Steps	Expected	Result
SEC-01	User U1 only in Access Group with D1	Device list = D1 (+ DeviceUser devices)	
SEC-02	U1 opens AI Analytics / Energy for D1	Data loads	
SEC-03	U1 requests sensor history for foreign device ID (DevTools)	`403` or `404`	
SEC-04	U1 opens Anomalies	Only alarms for accessible devices	
SEC-05	Org Admin still sees all org devices	Unrestricted within org	
SEC-06	User cannot PUT `/custom-dashboards/power-flow`	`403`	
SEC-07	User cannot POST `/access-groups`	`403`	

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13. API smoke tests (optional)

Replace `TOKEN` and IDs. Base: `http://localhost:5000/api` (or your prefix).

```
# Login
curl -s -X POST "$API/auth/login" -H "Content-Type: application/json" \
  -d '{"email":"orgadmin@ems.com","password":"Admin@123456"}'

# Access groups
curl -s "$API/access-groups?limit=20" -H "Authorization: Bearer $TOKEN"

# Device groups
curl -s "$API/device-groups?limit=20" -H "Authorization: Bearer $TOKEN"

# Facilities
curl -s "$API/facilities" -H "Authorization: Bearer $TOKEN"

# Custom dashboards
curl -s "$API/custom-dashboards?limit=20" -H "Authorization: Bearer $TOKEN"

# Power flow (live loads)
curl -s "$API/custom-dashboards/power-flow" -H "Authorization: Bearer $TOKEN"

# Year aggregate
curl -s "$API/sensor-data/aggregate?deviceId=$D1&variableName=ActivePower&timeRange=365d" \
  -H "Authorization: Bearer $TOKEN"
```

ID	Check	Expected	Result
API-01	Login returns token	`success`	
API-02	Power flow JSON has `groups[].load` / `loadKw`	Numbers	
API-03	Facilities nodes include `deviceIds` after linking	Array	
API-04	`365d` aggregate	`200` + data array	
API-05	User token + forbidden device	`403`	

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14. Role matrix checklist

Capability	Super Admin	Org Admin	User	Pass?
Access Groups CRUD	☐	☐	✗ nav	
Device Groups CRUD	☐	☐	✗ nav	
Facility edit + device link	—	☐	✗	
Power Flow view	—	☐	✗	
Power Flow edit sources	☐/☐	☐	✗	
Custom dashboards CRUD	☐	☐	☐	

Capability	Super Admin	Org Admin	User	Pass?
See shared dashboards	☐	☐	☐	
Live widgets	☐	☐	☐	
ACL-limited devices	—	—	☐	

15. Regression / negative cases

ID	Steps	Expected	Result
NEG-01	Create access group with empty name	Validation error	
NEG-02	Save facility with no levels	Harmless empty / org-only tree	
NEG-03	Open dashboard widget with no device + no org devices	Empty state message, not crash	
NEG-04	Break backend; reload Power Flow	Error UI / retry, no white screen	
NEG-05	Rapid create/delete dashboards	No duplicate key UI errors	
NEG-06	Two browsers: Org edits access group; User refreshes	Membership takes effect	

16. Sign-off

Area	Pass	Fail	Blocked	Notes
Access Groups				
Device Groups				
Facilities				
Power Flow				
Custom Dashboards				
Live widgets				
ACL / security				
API smoke				

Tester signature: ____________________ **Date:** __________

Known issues / follow-ups:

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17. Troubleshooting

Symptom	Likely cause	What to do
Power Flow loads always `0`	No Redis latest / no `ActivePower`	Check ingest; Redis; device config variables
Widgets spin forever	API error / CORS / wrong base URL	Network tab; fix `VITE_` API URL
User sees all org devices	Not using ACL build / old backend	Restart backend; confirm `deviceAccess.js` deployed
Facility links lost after save	Hierarchy rebuilt without save of links	Save after linking; avoid renaming mid-link without save
Year chart empty	No long-history data	Accept empty; confirm `365d` not 400
Access group save fails	CF tables missing	Run `prisma/add_cf_features.sql`
`403` on facilities replace	Logged in as User	Use Org Admin
PDF/docs outdated	N/A for app test	Ignore for E2E app QA

Recommended demo order (happy path)

1. Org Admin: Device Groups → Facilities + link devices
2. Org Admin: Access Group for User + D1
3. Org Admin: Power Flow on `/org`
4. Org Admin: Custom dashboard with live widgets
5. User: confirm device ACL + shared dashboard
6. Optional API curls for power-flow + aggregate

Appendix A — Quick curl login helper

```
export API=http://localhost:5000/api
export TOKEN=$(curl -s -X POST "$API/auth/login" \
  -H "Content-Type: application/json" \
  -d '{"email":"orgadmin@ems.com","password":"Admin@123456"}' \
  | sed -n 's/.*"accessToken":"\([^"]*\)".*/\1/p')
echo "TOKEN length: ${#TOKEN}"
```

> Adjust JSON field name (`accessToken` / `token`) to match your auth response shape.

Appendix B — Minimal telemetry check

On device detail or Data Center, confirm recent variables:

- `ActivePower` (kW) — Power Flow + many widgets
- `PowerConsumption` — energy / cost / carbon
- `ExportPower` — solar source derivation

Without these, UI tests can still Pass for structure; mark live-value cases **Blocked**.